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Started	Wednesday, 28 October 2026, 7:55 PM
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Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 80,150,200,470,40,90,300,400,70,120
- b. 200,470,80,150,40,90,300,400,70,120
- c. 40,80,90,150,200,300,400,470,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It builds a sorted array one element at a time
- b. It compares adjacent elements and swaps them if necessary
- c. It repeatedly finds the minimum element and moves it to the sorted part of the list
- d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **3**
Complete
Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Linear Search
- c. Merge Sort
- d. Binary Search

The correct answer is: Merge Sort

Question **4**
Complete
Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Insertion sort
- b. Heap sort
- c. Quick sort
- d. Bubble sort

The correct answer is:
Insertion sort

Question **5**
Complete
Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Extraction
- c. Distribution
- d. Selection

The correct answer is:
Insertion

Question **6**
Complete
Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. The parts are ignored
- b. Each part is sorted independently
- c. Each part is searched for a specific element
- d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **7**
Complete
Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Rearranging
- b. Sorting
- c. Searching
- d. Merging

The correct answer is: Sorting

Question **8**
Complete
Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Selection
- b. Searching
- c. All of the above
- d. Duplicates

The correct answer is: All of the above

Question **9**
Complete
Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Sorting data sequentially
- c. Sorting data in a single pass
- d. Rearranging data without sorting

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **10**
Complete
Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It has no effect on the search operation
- c. It makes the search operation slower
- d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching

Question **11**
Complete
Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Sorting
- b. Merging
- c. Complexity
- d. Searching

The correct answer is:
Complexity

Question **12**
Complete
Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Insertion Sort
- b. Merge Sort
- c. Bubble Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **13**
Complete
Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Binary Search
- b. Linear Search
- c. Bubble Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **14**
Complete
Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Bubble Sort
- b. Merge Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Merge Sort

Question **15**
Complete
Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It slows down the process
- b. It makes the data unsorted
- c. It makes it easier to select items based on their relationship to the rest of the items
- d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items